

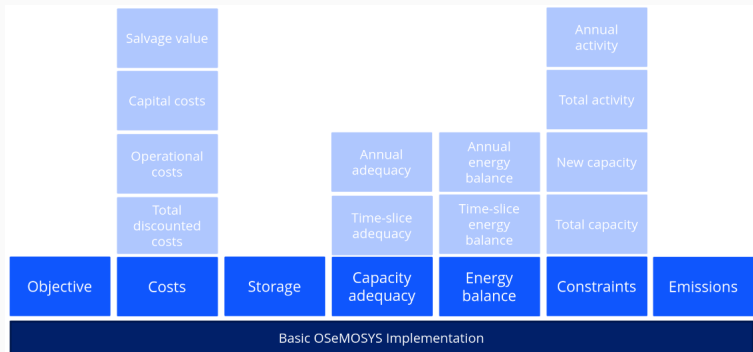
Introduction to OSeMOSYS

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OSeMOSYS is...

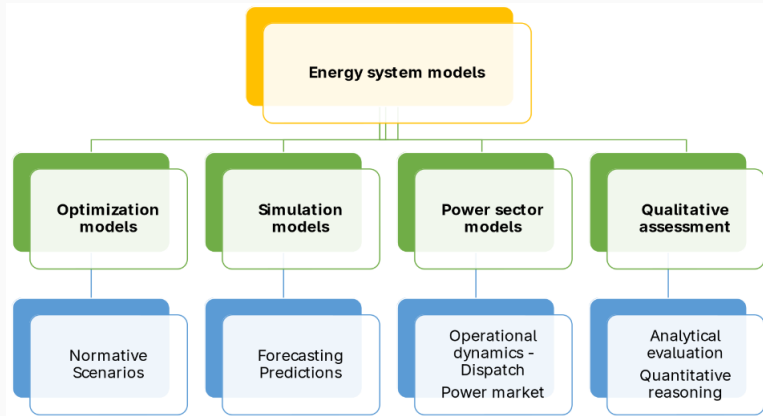
An open-source software used for energy modeling system and capacity expansion planning.



Energy System Models

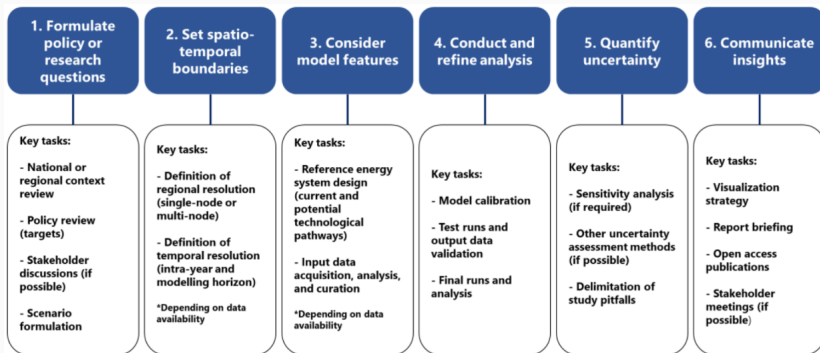
1. Models are simplified representations of reality
2. Energy systems can be simplified and represented in organized model structures
3. Energy system models aim to represent all components and process in energy systems with mathematical equations
4. This allows different future scenarios to be assessed
5. Crucially, energy system models provide useful insights, but cannot predict the future with certainty.

Energy System Models



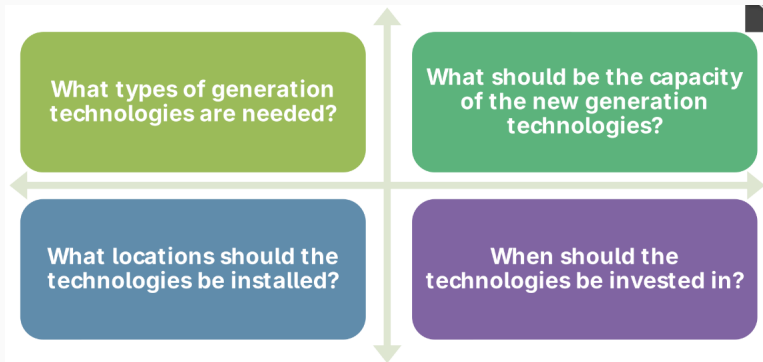
Energy System Models: Example...

This is the example of an energy system modeling for a pipeline.



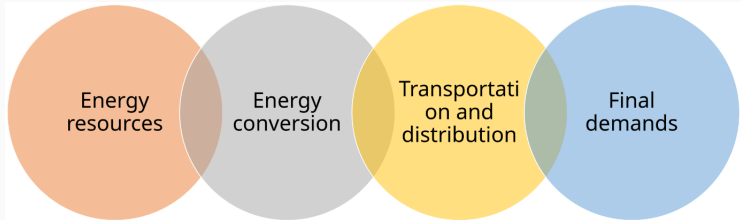
Energy System Models

Four outcomes the energy modeling.



Energy System Modeling

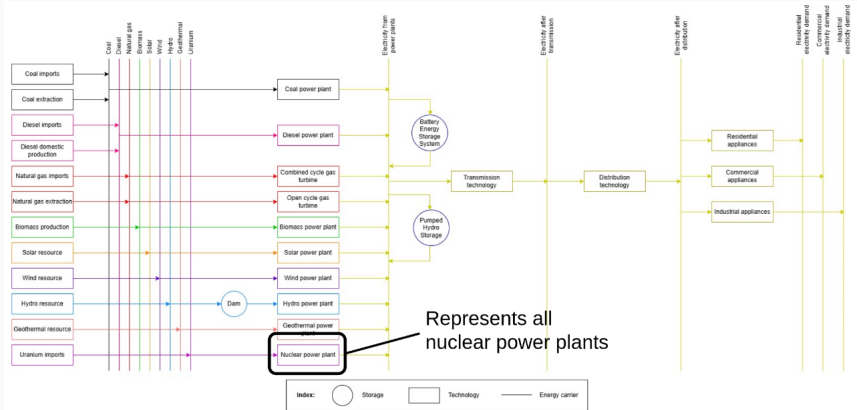
Energy systems include everything from full production to final energy consumption



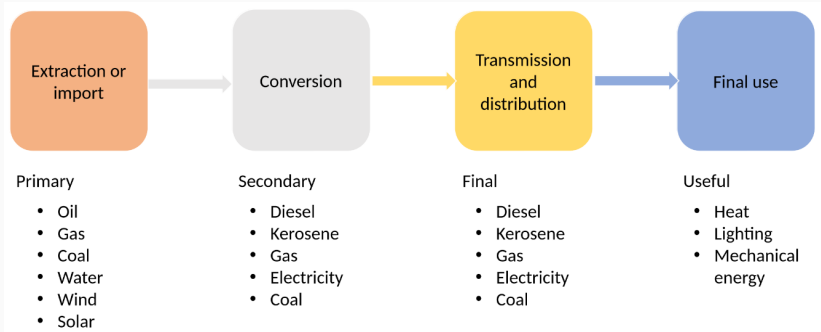
There are

1. Technology: Supply/produce a resource; convert one commodity into another.
2. commodity: Flows between technologies and final demands.

ESM: Components

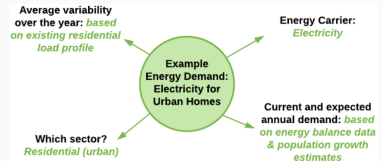
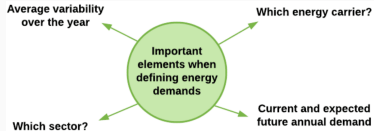


Reference Energy System



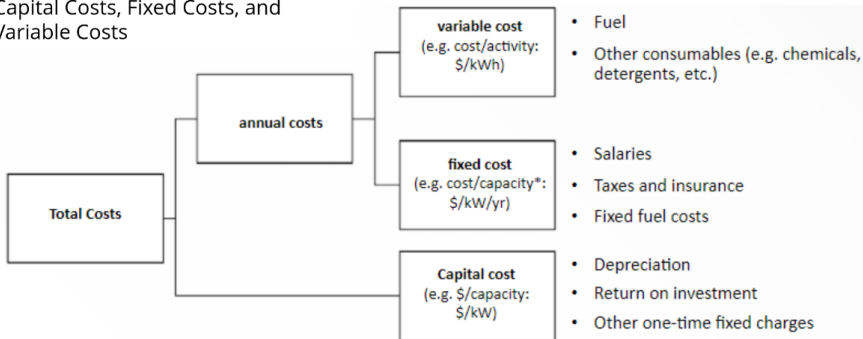
Demands in Modeling

Important elements in energy demands.



Costs in OSeMOSYS

In OSeMOSYS, costs are split into Capital Costs, Fixed Costs, and Variable Costs

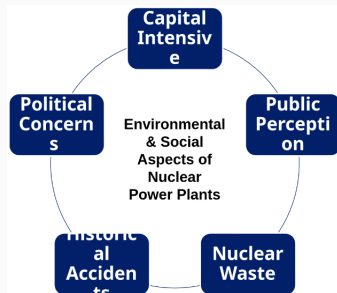
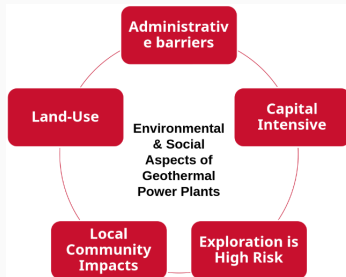
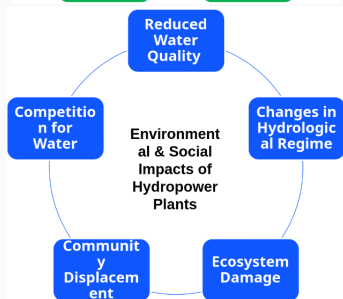
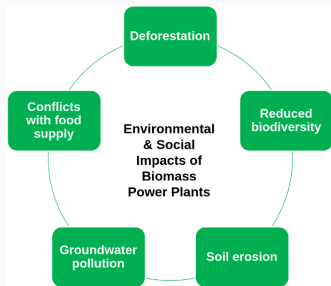


Examples onshore and offshore wind power plant.

1. Capital Cost: \$1880/kW
2. Fixed Cost: \$48/kW
3. Operational Life: 25 years

1. capital Cost: \$4700/kW
2. Fixed Cost: \$165/kW
3. Operational Life: 25 years

Environment and Social Impacts



Energy Storage Modeling

There are 6 different energy storage systems modelings

Thermal energy storage

Ex: Sensible heat storage using molten salts

Mechanical energy storage

Ex: Pumped hydro energy storage

Chemical energy storage

Ex: Hydrogen energy storage

Electrochemical energy storage

Ex: Battery energy storage

Electrical-magnetical energy storage

Ex: Supercapacitors

Others

Ex: Hybrid energy storage

Defining the Input Activity Ratio (IAR) and Output Activity Ratio (OAR) for the power technology(ies) follows a similar process to conventional technologies. The main difference lies in the definition of the OAR.

Input Activity Ratio is the inverse of charging efficiency using Technology to Storage equal to 1.

$$IAR = \frac{1}{\eta_{charge}}$$



Output Activity Ratio is the inverse of discharging efficiency using Technology from Storage equal to 1.

$$OAR = \frac{1}{\eta_{discharge}}$$

